INCH-POUND

MIL-DTL-50523A
2 August 2011
SUPERSEDING
MIL-C-50523(AR)
8 May 1970

DETAIL SPECIFICATION

CHARGE, DEMOLITION - M112
PARTS FOR LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts for one type of demolition charge designated as charge, demolition - M112 parts for loading, assembling and packing.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1375

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117	-	Bags, Heat Sealable

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Method for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC) DRAWINGS

8876127	-	Box, Wirebound, Packing, Ammunition for Charge, Demolition, M112
9204248	-	Charge, Demolition, M112

(Copies of these drawings are available online at pica.drawing.request@conus.army.mil or from US Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel

(Copies of the above document can be obtained from the ASTM headquarters, 100 Barr Harbor Drive, West Conshohocken, PA 19428 or online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.4.

3.2 M112 demolition charge, components and assemblies.

3.2.1 Demolition charge. The demolition charge shall conform to drawing 9204248.

3.2.1.1 Physical properties. Each individual charge will have a force resistance to the penetrameter of not less than 2.0 pounds and not greater than 8.0 pounds when tested.

3.2.1.2 Specific gravity. The specific gravity of the charge shall be not less than 1.50.

3.2.1.3 Energy output. The demolition charge shall cut a 1.0 inch thick steel witness plate, ASTM A36 grade 1018, completely into two separate sections.

3.2.1.4 Ability to cut and mold the charge. The charge shall not break, crumble, or otherwise lose its shape when cut or handled.

3.2.2 Wirebound box. The wirebound box shall conform to drawing 8876127.

3.2.2.1 Heat seal of barrier bag. The barrier bag heat seal shall exhibit no separation for the duration of 5 minutes.

3.3 Ammunition lot number. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Workmanship. Parts and assemblies shall be free of cracks, tears, voids, burrs, chips, dirt, grease, rust corrosion of packaging and metal clip-tie, and other foreign material that may adversely affect appearance or be injurious to personnel handling the item. The demolition charges shall be clean and free of holes with the exception of those due to force gage physical property testing of the penetrameter when performed in accordance with 4.5.1. The cleaning method used shall not be injurious to any part or assembly nor shall the cleaning agent contaminate them. All required markings shall be neat, separated and sharply defined. All packing materials must be dry at time of packing.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference.

Method of verification		Classes of verification						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First article B – Conformance						
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article		X	X	X	X		4.3
3.1.2	Conformance inspection		X	X	X		X	4.4
3.2.1	Demolition charge			X	X	X	X	4.4.2.1-4.4.2.4
3.2.1.1	Physical properties				X	X	X	4.5.1
3.2.1.2	Specific gravity				X	X	X	4.5.2
3.2.1.3	Energy output				X	X	X	4.5.3
3.2.1.4	Ability to cut and mold charge		X			X	X	4.5.1, 4.5.3
3.2.2	Wirebound box			X	X	X	X	4.4.2.2 - 4.4.2.4
3.2.2.1	Heat seal of barrier bag				X	X	X	4.5.4
3.3	Ammunition lot number			X		X	X	4.5.5
3.4	Workmanship			X		X	X	4.4.2.1-4.4.2.4

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Physical configuration audit (see paragraph 4.2)
b. First article (see paragraph 4.3)
c. Conformance (see paragraph 4.4)

4.2 Physical configuration audit (PCA). When specified, a sample shall be submitted to Physical Configuration Audit (PCA).

4.2.1 PCA quantity. The PCA shall consist of 100% verification of all requirements of drawings 13021902 and all of their components, assemblies, subassemblies, materials and components. Not less than five (5) samples of each component, etc. for all requirements shall be verified.

4.2.2 PCA rejection. If any item of the sample fails to comply with the PCA requirements, the sample shall be rejected.

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4.3 First article. A sample shall be subjected to first article verification in accordance with Table II.

4.3.1 First article quantity. First article inspection shall be performed on the quantities listed in Table II.

4.3.2 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected.

TABLE II. First article inspection.

Examination or Test	No. of Sample Units	Requirement Paragraph	Inspection Method Reference
Examination for defects	29	3.2.1, 3.2.2	4.4.2
Physical properties	29	3.2.1.1	4.5.1
Specific gravity test	10	3.2.1.2	4.5.2
Energy output test	5	3.2.1.3	4.5.3

4.4 Conformance inspection.

4.4.1 Lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916, paragraph “Acceptance by tables.” In addition, each inspection lot of charges shall contain:

a. Composition C-4 of not greater than one lot interfix number from one manufacturer.
b. Not greater than 50,000 M112 charges of one lot interfix number from one manufacturer.
c. Charges from not more than three consecutive weeks’ production.
d. Re-extrusion of up to ten percent of the M112 charges in a lot is a rework action that can be performed until Lot Acceptance Testing (LAT). Re-extrusion of more than ten percent of the M112 charges in a lot or re-extrusion after LAT shall require approval.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in the conformance examination/test

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paragraph and MIL-STD-1916. Unless specified otherwise, Verification Level IV shall be used for all characteristics defined as majors and Verification Level II for all minor characteristics.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraphs 4.4.2.1 thru 4.4.2.4. For all conformance inspections the same sample specimen may be used for non-destructive examinations or tests.

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4.4.2.1	M112 demolition charge.			Drawing Number 9204248
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly / Inspection Method Reference
Critical	None defined			
Major				
101	Weight	VL-IV	3.2.1	Balance
102	Length	VL-IV	3.2.1	Gage
103	Width	VL-IV	3.2.1	Gage
104	Thickness	VL-IV	3.2.1	Gage
105	Physical Properties	1/	3.2.2	4.5.1
106	Specific Gravity	2/	3.2.3	4.5.2
107	Energy Output Test	3/	3.2.4	4.5.3
108	Charge container missing, punctured, torn or otherwise damaged exposing explosive	VL-IV	3.2.1	Visual
109		VL-IV	3.2.1	Gage
110	Width of heat seal	VL-IV	3.2.1	Visual and Gage
111	Pressure sensitive tape missing, improperly applied, or incorrect length	VL-IV	3.2.1	Visual
112	Marking (other than warning marking) missing, incorrect or unidentifiable	VL-IV	3.2.1	Visual and Tactile
113	Sealing clip missing or insecure Warning marking missing, illegible or incorrect	100%(b)	3.2.1	Visual
Minor				
201	Charge improperly assembled into container	VL-II	3.2.1	Visual
202	Wax paper not removed (when applicable)	VL-II	3.2.1	Visual
203		VL-II	3.3	4.5.5
204	Ammunition lot numbering Evidence of poor workmanship	VL-II	3.4	Visual

Notes:
1/ Table III, MIL-STD-1916, VL IV. Reduced inspection level not authorized.
2/ Total of 10 sample units.
3/ Total of 5 sample units.
(b) 200% shall be required for manual visual inspections. 100% is acceptable only if inspection is being performed with an automated vision system.

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4.4.2.2	Wirebound packing box prior to sealing barrier bag.			Drawing Number 8876127
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly / Inspection Method Reference
Critical	None Defined			
Major				
101	Box improperly assembled	VL-IV	3.2.2	Visual
102	Incorrect number of demolition charges	VL-IV	3.2.2	Visual
103	Barrier bag punctured	VL-IV	3.2.2	Visual
Minor				
201	Filler packing material missing	VL-II	3.2.2	Visual
202	Charges incorrectly packed	VL-II	3.2.2	Visual
203	Evidence of poor workmanship	VL-II	3.4	Visual

4.4.2.3	Wirebound packing box prior to sealing.			Drawing Number 8876127
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly / Inspection Method Reference
Critical	None Defined			
Major				
101	Barrier bag improperly sealed	VL-IV	3.2.2	Visual
102	Heat seal test of barrier bag	1/	3.2.2.1	4.5.4
Minor				
201	Packing material missing	VL-II	3.2.2	Visual
202	Evidence of poor workmanship	VL-II	3.4	Visual

Notes: 1/ Total of 5 sample units.

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4.4.2.4	Wirebound packing box sealed.			Drawing Number 8876127
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Next Higher Assembly Inspection Method Reference
Critical	None defined			
Major				
101	Loops improperly closed	VL-IV	3.2.2	Visual
102	Wire broken	VL-IV	3.2.2	Visual
103	Box damaged	VL-IV	3.2.2	Visual
Minor				
201	Contents loose	VL-II	3.2.2	Visual
202	Markings misleading or unidentifiable	VL-II	3.2.2	Visual
203	Box seal missing, unsealed, or improperly positioned	VL-II	3.2.2	Visual
204	Evidence of poor workmanship	VL-II	3.4	Visual

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4.5 Testing.

4.5.1 Physical properties. (see 6.6) This test measures the force required for a cone point to penetrate the M112 demolition charge for a specified distance. The samples shall be selected after the charges have been packaged in charge containers. Any charge that breaks or physically separates during the sample preparation will not be replaced. The breakage or separation shall be a nonconforming unit and result in lot rejection. External temperature of the charges at time of test shall be 73 +/- 3 degrees F (22.8 +/- 1.7 degrees C). Samples shall be temperature conditioned for a minimum of 24 hours prior to removing from the charge container. Each sample demolition charge shall be removed from its charge container and two penetrameter readings shall be taken on each M112 charge at approximately one inch from each end of the sample charge. The maximum of the two penetrameter readings shall be recorded unless one reading is less than 2.0 pounds or more than 8.0 pounds. If one reading is below 2.0 or greater than 8.0, that reading will be recorded and the demolition charge shall be considered a nonconforming unit. The lot shall be considered acceptable if the inspection sample contains no nonconforming units and the applicable “k” criteria are met. Both the cone point and the M112 charges shall be on solid supports. The design and dimensions of the cone point shall be as specified in Figure 1. As the cone point comes into contact with the M112 charge, the penetrameter load gage housing will continue traveling toward the charge at a constant speed of 6 +/- 1/8 inches per minute. After the housing has traveled 0.25 +/- 0.06 inches, the penetrameter load gage housing shall stop. This is a non-destructive test. Tested samples may be rewrapped and returned to the lot.

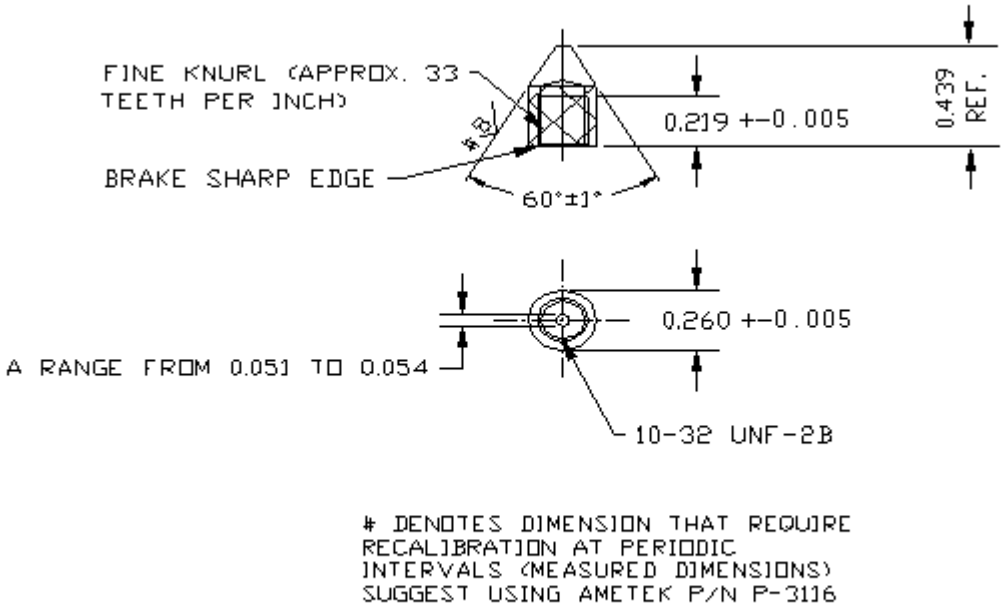


Figure 1. Penetrameter cone point specifications (dimensions in inches).

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4.5.2 Specific gravity. This is a destructive test; samples cannot be returned to the lot. The sample shall be selected randomly after the charges have been packaged in charge containers. External temperature of the charges at time of test shall be 73 ± 3°F (22.8 ± 1.7°C). Samples shall be temperature conditioned for a minimum of 24 hours prior to removal from charge container. The specific gravity of each sample M112 charge shall be determined as follows:

a. Remove each demolition charge from its charge container being careful not to damage the charges.
b. Weigh the (dry) charge to the nearest .01 pound.
c. Weigh the charge immersed in water with water temperature at 77 +/- 9 degrees F (25 +/- 5 degrees C), taking care to remove all air bubbles remaining on the surfaces of the charge.

$$\textit{Specific gravity} = \frac{\textit{Weight of charge dry}}{\textit{(Weight of charge dry minus weight of charge immersed in water)}} \times \textit{Specific gravity of } H_2O$$

d. Failure of any one of the sample demolition charges to meet a minimum specific gravity of 1.50 shall result in lot rejection.

4.5.3 Energy output test. Sample shall be randomly selected from each M112 lot after the charges have been packaged in charge containers. External temperature of the charges at time of test shall be 73 +/- 3 degrees F (22.8 +/- 1.7 degrees C). Samples shall be temperature conditioned for a minimum of 24 hours prior to removal from charge container. The M112 demolition charge assembly shall be measured against a steel witness plate to verify effectiveness of the explosive.

4.5.3.1 Effectiveness of explosive. The charge shall be cut from each M112 charge to the dimensions of 0.5 x 1.5 x. 10.0 inches (nominal). Any charge that breaks during the sample preparation will not be replaced. The breakage will be recorded as part of the test data. Charge will not be considered defective if it can be kneaded/hand pressed into the required shape. Failure of the charge to be kneaded/hand pressed into the required shape or failure to pass the energy output test will result in lot rejection. The steel witness plate is an ASTM A36 steel with dimensions of 1.0 x 10.0 x 10.0 inches (nominal). The charge shall be applied to the witness plate and primed at the center using either M6 or M7 blasting cap on top of the charge. The blasting cap end shall be covered with a strip of C-4 explosive approximately 0.5 x 1.0 x 4.0 inches. The primed area shall be secured with tape to ensure intimate contact between the charge and the witness plate (see Figure 2). Failure of any one of the sample demolition charges to cut a steel witness plate completely into two separate sections will result in lot rejection.

4.5.3.2 Witness plate setup. Each witness plate shall be placed on two (2) solid steel supports with a 4 inch span between the supports as shown in Figure 2.

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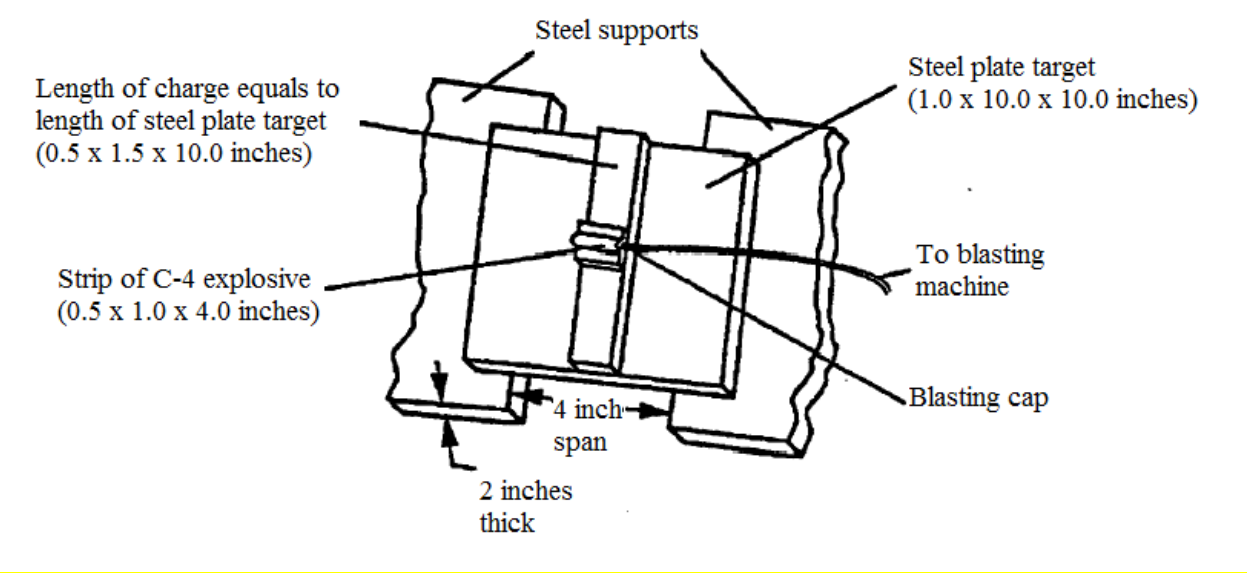


Figure 2. Energy output test set up.

4.5.4 Heat seal test of barrier bag. A sample of packed and sealed barrier bags shall be randomly selected from each lot. The test shall be performed as specified in MIL-STD-2073-1 or MIL-DTL-117. If any sealed barrier bag fails to comply with the requirement specified, the lot shall be rejected.

4.5.5 Ammunition lot numbering. Visually verify that an ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contractual document or order. When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Block Demolition Charge M112 is plastic explosive. This charge is ideally suited for cutting charges as the adhesive backing allows the charge to be attached to any relatively flat, dry surface. The explosive may also be cut and/or removed from the Mylar wrapper and hand formed as desired to suit the target.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Requirements for Ammunition Lot Numbering, if applicable.

d. Requirement for acceptance inspection equipment (AIE) designs.

e. Certificate of Conformance for each lot or shipment of product.

f. Requirements for submission of explosive description sheets in accordance with MIL-STD-1171.

g. Preservation, packaging and marking:

Level A. The demolition charges should be preserved and packaged in accordance with drawing 8876127. Marking should be in accordance with drawing 8876127.

h. The National Stock Number and Department of Defense Identification Code for MIL-DTL-50523, Charge, Demolition: M112 is 1375-01-389-3854-M023.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required per the Contract Data Requirements Lists (DD Form 1423).

6.4 Distribution of ammunition data cards. The distribution of data cards should be made in accordance with the requirements furnished in the contractual document.

6.5 Performance oriented packaging (POP). Prior to shipment, the contractor should make sure the container has been tested by a government approved Performance Oriented Packaging (POP) Test Laboratory for compliance with POP requirements in accordance with Title 49 Code of Federal Regulation. Test is to a weight at least 10% greater than the applicable gross weight (based on previous production or the estimated gross weight shown on the item Packing and Marking Drawing). POP marking should not be applied to the container until verified by the government. The POP test report should be generated by the contractor (DI-PACK-81059) (Data Item Description) following the test. The report should be kept on file by the contractor and should also be submitted (DI-PACK-81059) to the U.S. Army Tank-automotive and Armaments Command - Armament Research, Development and Engineering Center, ATTN: RDAR-EIL-P, Picatinny Arsenal, New Jersey, 07806-5000. For multi-year contractual documents, the contractor should re-perform POP testing at a certified test laboratory if: (a) the initial POP test report expires before the end of the contract or (b) there is a change in container manufacturer or design of the exterior shipping container. No re-test is needed if all packaging is purchased while under an unexpired POP test.

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6.6 Inspection equipment designs. Inspection equipment designs are of two types – Government Special Inspection Equipment (SIE) and contractor designs. The SIE designs are designated by drawing numbers under the "Conformance Inspection" headings in section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with paragraphs 6.6.2 and 6.6.3. The contractor should correctly utilize inspection equipment and be charged with the responsibility of insuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes.

6.6.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawing which completely depicts all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in ASME Y14.100.

c. An envelope drawing, as defined in ASME Y14.100, which establishes the criteria which a detailed design should meet. When envelope drawings are specified, the contractor should prepare designs which comply with the criteria therein.

6.6.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposed to use. (Commercial equipment is defined as unmodified equipment that is cataloged and available for purchase by the general public.) Contractor designs should include appropriate operating instructions, calibration procedures, and maintenance procedures. Commercial equipment should be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs should be prepared for any special fixtures required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see paragraph 6.5.1c). Designs should be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Design of Inspection Equipment for Dimensional Characteristics" for guidance.

6.6.3 Submission of designs for Government approval. Designs should be submitted for approval in accordance with the Contract Data Requirements Lists (DD Form 1423) or in the contractual document. All submissions should include Reliability & Repeatability Studies of the inspection equipment consisting of data and methodology used.

6.7 Subject term (key word) listing.
Barrier Bag
Wirebound Box

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – AR
Navy – OS
Air Force – 11

Preparing activity:
Army – AR
(Project: 1375-2010-015)

Review activities:
Navy – AS, NP
Air Force – 70, 99

Civil agency:
GSA - FAS

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